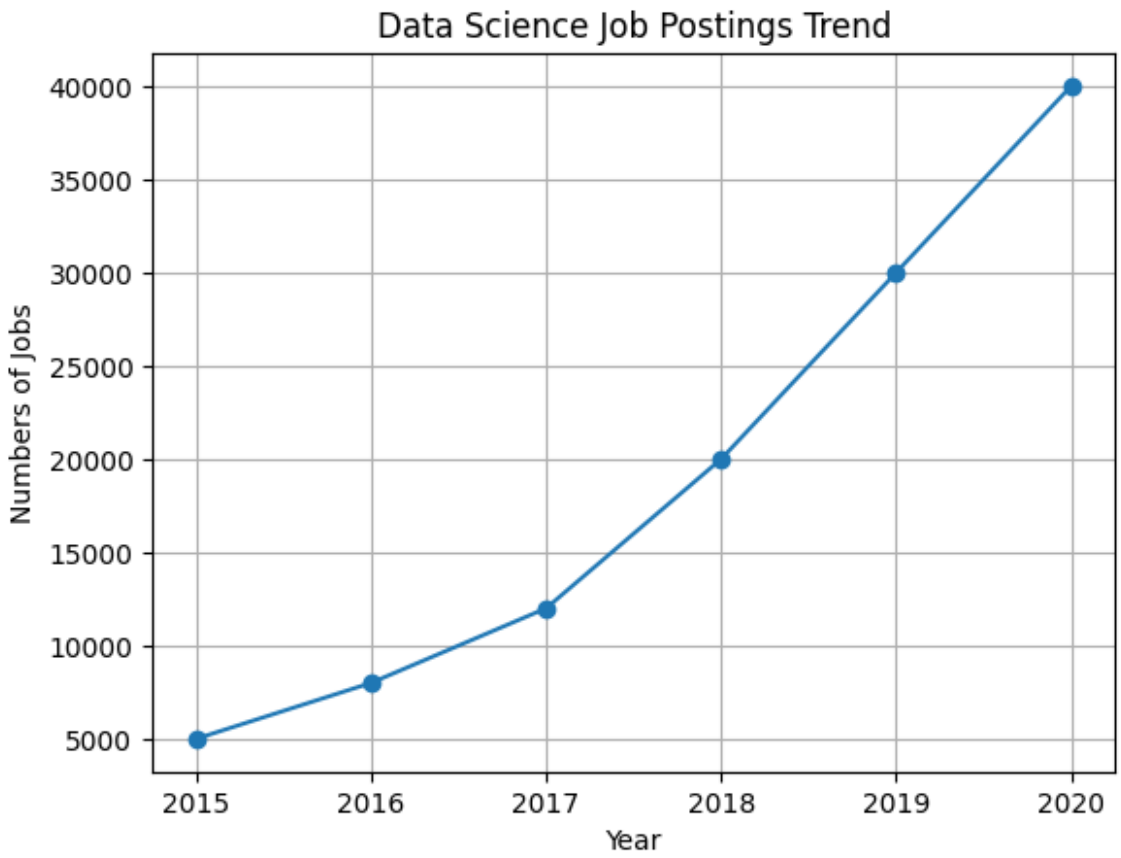
```python
import matplotlib.pyplot as plt
import pandas as pd

data = {
    "Year": [2015, 2016, 2017, 2018, 2019, 2020],
    "Jobs": [5000, 8000, 12000, 20000, 30000, 40000],
}

df = pd.DataFrame(data)
print(df)

plt.plot(df["Year"], df["Jobs"], marker="o")
plt.title("Data Science Job Postings Trend")
plt.xlabel("Year")
plt.ylabel("Numbers of Jobs")
plt.grid()
plt.show()
```

```
   Year   Jobs
0  2015   5000
1  2016   8000
2  2017  12000
3  2018  20000
4  2019  30000
5  2020  40000
```




```
import matplotlib.pyplot as plt
import pandas as pd

import pandas as pd

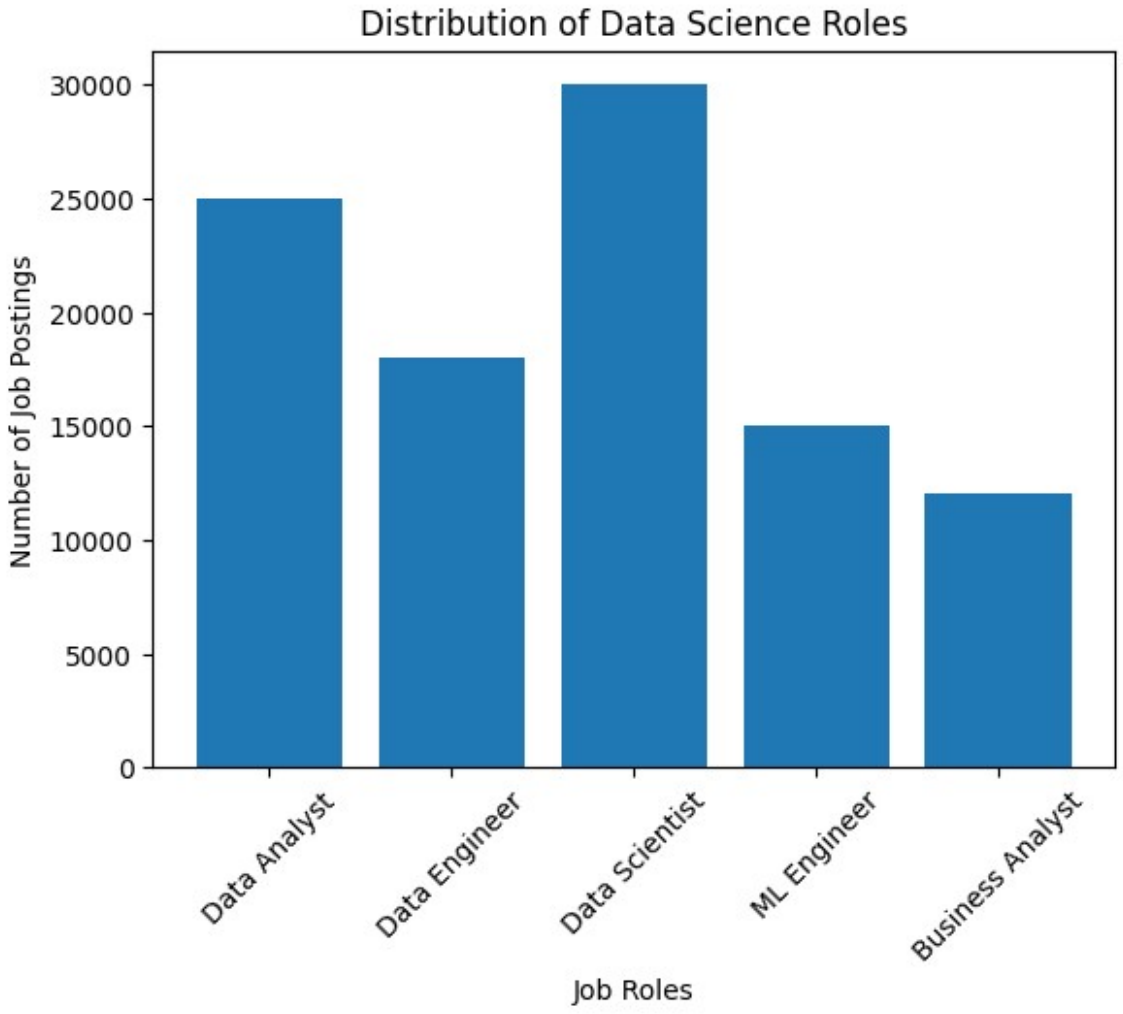
data = {
    "Role": [
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "ML Engineer",
        "Business Analyst",
    ],
    "Number_of_Jobs": [25000, 18000, 30000, 15000, 12000],
}

df = pd.DataFrame(data)
print(df)

plt.bar(df["Role"], df["Number_of_Jobs"])
plt.title("Distribution of Data Science Roles")
plt.xlabel("Job Roles")
plt.ylabel("Number of Job Postings")
```

```
plt.xticks(rotation=45)
plt.show()
```

| | Role | Number_of_Jobs |
|---|---|---|
| 0 | Data Analyst | 25000 |
| 1 | Data Engineer | 18000 |
| 2 | Data Scientist | 30000 |
| 3 | ML Engineer | 15000 |
| 4 | Business Analyst | 12000 |




```
import pandas as pd

# Structured Data
structured = pd.DataFrame({"Name": ["John", "Alice"], "Age": [25,
30]})

print("Structured Data")
print(structured)
```

```python
# Semi-structured Data
semi_structured = {"Name": "Bob", "Age": 28, "Skill": "Python"}

print("\nSemi-structured Data:")
print(semi_structured)

# Unstructured Data
unstructured = "Data Science uses machine learning and AI"

print("\nUnstructured Data:")
print(unstructured)
```

```
Structured Data
    Name  Age
0   John   25
1  Alice   30

Semi-structured Data:
{'Name': 'Bob', 'Age': 28, 'Skill': 'Python'}

Unstructured Data:
Data Science uses machine learning and AI
```

```python
!pip install cryptography
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)

data = "My password is 12345"

encrypted_data = cipher.encrypt(data.encode())
print("Encrypted data:")
print(encrypted_data)

decrypted_data = cipher.decrypt(encrypted_data).decode()

print("\nDecrypted Data:")
print(decrypted_data)
```

```
Requirement already satisfied: cryptography in
/usr/local/lib/python3.13/dist-packages (50.0.2)
Requirement already satisfied: cffi>=2.0.0 in
/usr/local/lib/python3.13/dist-packages (from cryptography) (2.1.1)
Requirement already satisfied: pycparser in
/usr/local/lib/python3.13/dist-packages (from cffi>=2.0.0-
>cryptography) (3.0)
Encrypted data:
b'gAAAAABqxnZwAyDox_NJHjhmvhzTiq5RCPLAD609HuActesq964Et3LchbsAaSa3OpVl
svc4vpIpfxzKSv8j1y4FtTRqOqsjvvp2uXIzHdHbunSBbxp2k6w='
```

```
Decrypted Data:
My password is 12345
```